One argument, several ways to read it

An architectural companion to Lean Exposition

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AI assistance. Will Cook directed the work. AI agents based on large language models produced and revised this manuscript.

A mathematician reading a short proof may want to stop at its main idea, check one difficult estimate, or learn the subject from the beginning. Those are different requests. A proof assistant can check a precisely stated theorem, but its successful check does not choose an explanation for any of these readers. Nor does a readable explanation, by itself, show how faithfully it represents the checked statement.

Lean Exposition connects three written accounts of selected mathematics to a shared record of claims, assumptions, sources and checks. Each account has its own purpose and reading order. Their connections let a reader leave an explanation at the point of doubt, recover the relevant detail, and return without losing track of which statement was being discussed. This companion explains that arrangement and the choices behind it. The mathematical papers supply the arguments themselves. The repository implements the records and reading tools described here; the worked journeys show what they preserve and what a reader or author still has to decide.

1. Why write more than one account?

The starting material may include an original research paper, a formal development in the Lean proof assistant, and earlier literature. These sources answer different questions. The research paper explains a mathematical argument in the author's chosen language. The Lean development gives precise definitions and formal proofs. Earlier literature establishes what was known, where an input came from and which conclusions require additional results. Reading one does not necessarily recover the content of the others.

An author first reconstructs the selected argument in a **long reasoning record**. This is where a reader should be able to find the derivation behind an estimate, the treatment of an endpoint, or the reason two formulations correspond. Its length follows the argument that needs recovering. It can retain a failed approach when the failure explains why a hypothesis matters. It also identifies steps that have not been justified.

A **short research paper** then selects a contribution worth presenting as a connected argument. It gives the statement, its position in the literature and the decisive proof, with precise destinations for substantial details it omits. The selection may leave whole further results in the long record. Those omissions should make the paper more focused without making its claims stronger.

An **expository paper** makes a different selection. It teaches a result to readers with stated

prerequisites, introducing objects through examples and explaining why each construction is needed. It may spend more time on a definition than the specialist paper, while covering fewer consequences. Its author keeps the original sources and detailed record available; the exposition need not be produced by repeatedly shortening the short paper.

The supplied example concerns the Sherrington–Kirkpatrick (SK) model, a random optimisation problem on interacting spins. Its exposition begins with a two-spin calculation and explains why an optimal function used to express the model's value cannot stay flat on an interval. The short paper uses one quantitative estimate twice: first to exclude these gaps, then in a further argument for positive density. The exposition teaches the contact argument by scaling weights introduced in an earlier example; the specialist account uses local removal of mass. The long record retains both proofs, reconstructs the analytic estimates and continues into the separate argument about the Quantum Approximate Optimization Algorithm (QAOA). A reader interested in the no-gap mechanism does not have to read the circuit construction first.

These are entrances for different purposes. The writing process gives authors more context than any one reader is asked to absorb.

2. What the shared graph remembers

Maintaining three accounts creates a practical problem. If a hypothesis is corrected in the detailed argument, which shorter passages need attention? If a reader doubts an estimate, where is it explained, and which source is being invoked? A bibliography alone cannot answer these questions at the level of a particular passage.

The project records selected claims and their relationships in a graph. A claim has a stated scope and associated assumptions. It can point to an ingredient used in its argument, an original source, or a formal statement with which the author proposes a correspondence. Sources retain their editions; checking records say what was checked and with what outcome. This is a curated, partial account of the available mathematics. It does not pretend to contain every declaration or every dependency in the underlying Lean development.

Each written paper has a **passage map**: annotations connecting selected passages to the claims, assumptions and citations they discuss. Thus two papers can refer to the same claim while explaining it differently. Their prose remains independently readable, and text without a recorded mapping remains visible as such.

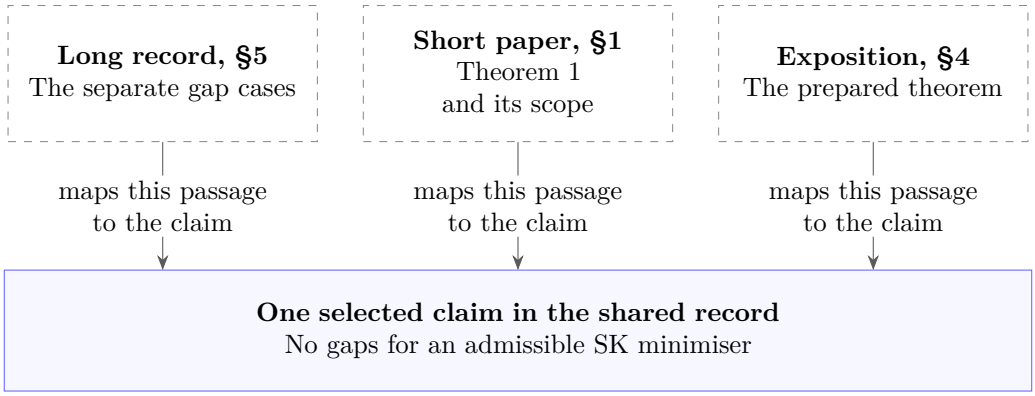


Figure 1: Three passages, one recorded claim. These passages directly name the same ordinary no-gap claim, whose hypotheses remain fixed. Their wording, proof choices and reading order can differ. The arrows denote authored passage mappings; they do not assert equivalent passages or automatic generation. Only one of the shared corpus's claims is shown.

An arrow needs a meaning. Here an argument arrow points from a claim to an ingredient it uses: following it asks what supports the claim. Following the same relation backwards asks what recorded arguments use that ingredient. The direction of the stored relationship stays the same. Several ingredients may be needed together; an arrow to one of them does not say that it alone proves the conclusion.

Other relationships answer other questions. A citation credits or refers to a source. A correspondence with a formal statement asks the reader to compare two formulations. A recorded Lean dependency says that a constant occurs in a declaration's formal type or proof, or in its structural references. A checking record identifies the source or claim it concerns. These relations must remain distinguishable: a source can be historically relevant without being a premise, and a proposed correspondence can require a mathematical argument of its own.

The order of paragraphs is another relationship again. An exposition may give an example before stating the general theorem; a specialist paper may state the theorem first. Neither order changes the theorem's prerequisites. A paper's reading route therefore consists of its authored sequence together with the support selected for its passages. The graph helps readers inspect that selection; it does not choose a sufficient proof or a good teaching order.

Local queries can also start from a source and find its recorded uses, or follow a claim back to its ingredients. They report when a paper restriction, exploration depth or result limit stops the search. The corpus is partial: an absent path is not a conclusion of mathematical independence. These boundaries matter when the same traversal is used to plan a correction.

3. Follow one claim back to its argument

The SK model asks which choices of signs maximise a random sum of pair interactions. The Parisi formula expresses its limiting optimum through a minimisation over functions called order parameters. In the pure model at zero external field, an admissible order parameter is a nonnegative, nondecreasing, right-continuous, integrable function $\gamma : [0,1) \to [0,\infty)$. Its associated measure records the increments of γ. A flat interval is a gap in those increments, even if the function's value there is positive.

The result at issue says that an admissible minimiser has no such gap: its measure has full support relative to $[0, 1)$. The endpoint one is excluded. Existence of a minimiser comes from a separate theorem of Auffinger and Chen. An earlier paper by Hong-Bin Chen established full support by another method. The worked accounts reconstruct the quantitative argument in OpenAI's full-support manuscript. Their contribution here is the explanation and its correspondence to sources.

Consider just one case: a bounded gap (ℓ, r) between support points, with $0 < \ell < r < 1$, on which γ has a positive constant value. To test whether γ could minimise, the proof constructs a differentiable curve q measuring the average squared response in an auxiliary optimisation problem. The mathematical papers explain that construction. Here we take two properties as inputs and follow how they exclude the gap.

Minimality gives **contact** at the endpoints. In the notation of the papers, this supplies

$$q(\ell) = \ell, \qquad q(r) = r, \qquad q'(\ell), q'(r) \leq 1.$$

The change in q across the interval must therefore be $r - \ell$. Separately, a strict-curvature estimate on a positive constant interval makes q' strictly convex. Since q' is continuous at the endpoints, its interior values lie strictly below their joining chord, which is at most one. Consequently

$$q(r) - q(\ell) = \int_{\ell}^{r} q'(t)\, dt < r - \ell.$$

The two requirements contradict each other. Neither endpoint contact nor strict curvature alone gives this contradiction.

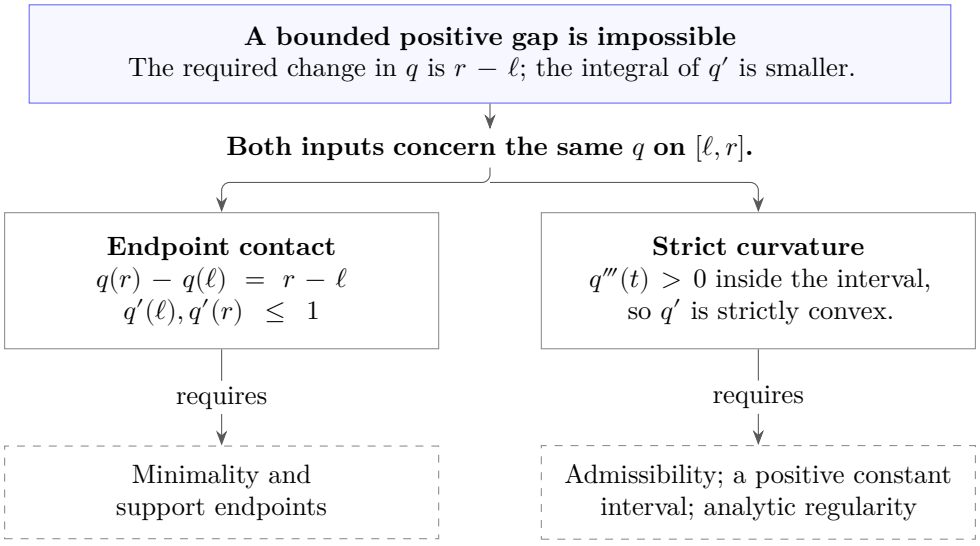


Figure 2: Which ingredients does the contradiction use? Arrows point from a conclusion to its inputs. Both branches are needed in the stated SK setting. Minimality supplies contact; the curvature estimate itself applies without minimality. This is one branch of the ordinary argument, not an extracted Lean proof graph. Zero coefficients and initial or terminal gaps require separate treatment.

Now suppose a reader accepts the final integral comparison but wants to know why the curvature is strict. The short paper, section "Minimality gives contact; curvature excludes gaps" identifies

the estimate at its use. Its next section, "The integrated estimate", gives the compact analytic explanation. Two long passages are recorded as expansions of that section. Long §§4.1–4.4 derive the integrated inequality: a correction with zero expectation makes positivity accessible even though the original integrand can be negative. Long §4.5 then transfers the estimate to the stated coefficient class and proves strictness for the limiting solution. Positive approximants alone would not suffice, since their margins could tend to zero.

These destinations answer different doubts. The first maps the analytic ingredients, while the second directly maps the strict-curvature claim. Both have an authored expansion link to the short treatment. A list of passages sharing the final claim would therefore miss one useful explanation. Figure 3 shows these links and the claim's citation of Proposition 5.1 of the original manuscript. The proposition applies on an admissible positive plateau without assuming minimality. The reader brings that conditional estimate back to the short account and resumes the gap proof: strict curvature puts q' below the endpoint chord, and contact makes its integral too small. The contact hypothesis has survived the excursion.

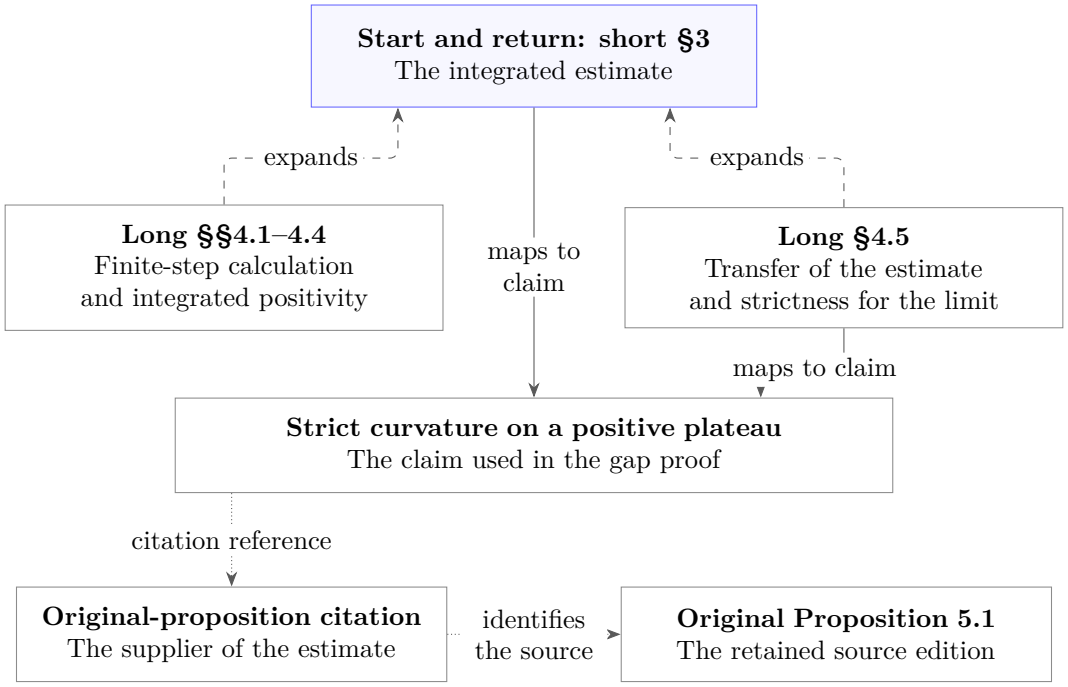


Figure 3: Leave an estimate, recover its argument, return. Dashed arrows record that a fuller passage expands the short treatment: a reader follows them backwards to visit the long record, then returns to the same short passage. Solid arrows are direct claim mappings. Dotted arrows follow the claim's citation to its source edition. The left-hand treatment supplies analytic ingredients without directly mapping the final curvature claim. These are selected reading and source relations, not proof edges.

The reading website presents authored links separately from shared-claim matches and can keep the original passage beside a visit to another account. A selected passage also opens its mapped claims and recorded inputs. The selection concerns whole passages, so it does not assign proof provenance to each selected sentence. In a standalone PDF, descriptive paper and section references preserve the route when the other files are unavailable; public source links remain clickable.

A reader who does not yet understand a gap can instead enter the exposition, section "What a gap means". Its example separates a positive cumulative value from vanishing increments. The later section "Why a flat interval cannot minimise" develops the contact and area argument. This route teaches the objects that the specialist route assumes. The graph connects the example to the Stieltjes-measure definition and records its use in the later argument; reading each passage supplies the explanation.

4. What survives a shorter account

Shortening the example cannot change "an admissible minimiser" into "an admissible function". Constant functions are allowed candidates, and the curvature estimate does not say they minimise the functional. Without minimality, the endpoint contact used in the contradiction is missing. The short account must retain that premise where it states and uses the no-gap conclusion.

It must also preserve the result's extent. Full support says that every nonempty relatively open interval receives mass. It does not alone assert a density, let alone a strictly positive density. The stronger density conclusion in the source has a further argument. Likewise, the broader QAOA claim requires approximation and circuit arguments beyond full support. Different accounts may stop at different conclusions, provided each makes its own stopping point clear.

The passage maps help enforce this discipline. For the claims an author selects, the checks require the recorded assumptions, citations and unresolved obligations to remain in the map. They can catch a missing reference or an unreconciled change to mapped text. They cannot tell whether the theorem's wording still states the assumption, or whether the cited argument is adequate. Those are questions for mathematical and editorial review.

Attribution survives the same selection. The no-gap theorem retains credit for the earlier support result and for the quantitative proof being explained. A reference may supply a theorem, an adapted construction or historical context. Recording that role avoids treating every citation as a logical dependency while still letting the reader recover the relevant source.

5. Explanation, formal evidence and navigation

A reader may next ask whether the no-gap conclusion has a formal counterpart. The project records a correspondence with a selected Lean statement. That statement assumes an admissible minimiser on a Brownian system and supplies full support, strict increase and associated moment identities. It does not assert minimiser existence or QAOA circuit convergence. The comparison specification states the target; a separate implementation source contains the candidate proof. Neither link certifies every sentence of the exposition.

Even a close correspondence may need explanation. The long record's section "The support statement and the derivative bridge" compares a formal limit along prescribed step sizes with the ordinary right derivative used in the manuscript. Convexity orders the intervening secant slopes, so their limit is fixed by the sampled ones. This mathematical bridge, with its hypotheses, is what connects the formulations. The correspondence record lets a reader locate it.

The design therefore preserves three distinct activities. The written account explains why a conclusion follows. Formal checking tests a precise statement and proof in a specified environment. Navigation helps a reader find the account, formal statement and records of the checks. None can stand in for the other two.

Checking outcomes also retain their scope. Identifying unchanged source bytes, building a Lean project, comparing it with a separately stated challenge, and reviewing its mathematical interpretation answer different questions. For the SK example, a source-bound Support build and dependency extraction completed on 26 October 2026. The extracted conditional full-support root retains its hypotheses and external dependency boundaries. Comparator acceptance and the scalar-target results remain pending. These observations do not certify the English accounts. Earlier unsuccessful runs remain observations of those runs, rather than counterexamples to the theorem.

Reviews concern particular editions. If an author changes a hypothesis or maps the same paragraph to a different formal statement, the correspondence has changed even when much of the prose has not. The project keeps the text, map and sources identifiable together so a review can be understood in its original context. A revised edition needs review of the affected claim.

6. How a correction changes the reading route

A recorded revision shows why the relationships between accounts need maintenance. An earlier exposition used an explicit calculation to exclude empty support, while placing that calculation in a section described as optional. The theorem was sound, but the promised main reading route still needed the later calculation. The revision record describes the decision to use a short qualitative argument instead. The current papers exhibit that change; this is a reconstruction from the recorded review and adoption, rather than an observed reader experiment.

Empty support would give $\gamma = 0$. The analytic inputs then make q strictly convex, with $q(0) = 0$ and $q(t) \le 1$ for $t < 1$. Here it is q itself that is convex, unlike q' in the positive-gap example. For $0 < t < s < 1$, a chord gives

$$q(t) \le \frac{t}{s} q(s) \le \frac{t}{s}.$$

Letting s tend to one yields only $q(t) \le t$. Strictness comes from a second, finite chord: for $t < v < 1$,

$$q(t) < \frac{t}{v} q(v) \le t.$$

Thus $\int_0^1 (q(t) - t)\, dt < 0$. Adding a small atom at zero is an admissible variation, and minimality requires this integral to be nonnegative. That contradiction closes the empty case without the explicit calculation or a value of q at one.

The same repair has different consequences in the three accounts. The long record, §5, retains the complete deduction. The short paper includes it in the gap-case proof. The exposition explains the second chord before its optional material. The exact zero-coefficient calculation and the stronger quantitative terminal estimate remain available for their additional information. Figure 2's bounded positive-gap argument is unchanged. Reconciliation means deciding which uses need which changes, rather than copying one replacement paragraph everywhere.

The new ordinary deduction has its own claim record and mappings to these three uses. Following recorded consumers gives an author candidates to inspect. It does not find every sentence affected by a reading promise: headings, optional routes, this companion and the guides also need reading. The current passage maps cover the three mathematical accounts, not every explanation in the repository.

Checks help with a different part of the repair. A reading link binds both passages to their editions. Changing its destination can therefore make the link stale even when its other end is unchanged.

Refreshing those bindings identifies the successor; it does not renew the old review. The author must judge the revised claim and the usefulness of the link, retaining the earlier review with the edition it concerned. A passage can also stay textually unchanged while its proposed correspondence changes.

The local editorial decision applied the existing requirement that an optional section be dispensable for the promised route. It supplies no causal evidence that a new writing rule improved the result. Written methods guide source selection, detailed reconstruction, expert selection and exposition; a criticism may call for applying existing guidance, moving it to the relevant decision, or changing it. A proposed general change needs checks on two different cases and an exception. For instance, replacing a quantitative lemma by an existence proof is harmless only if no retained use needs its bound. A source-supported local repair and a useful method for another author remain different achievements.

7. When is the extra structure useful?

Ordinary papers with careful references remain useful. The additional work here is maintaining explicit relationships across several accounts, so that a reader or an author can find a claim's assumptions, locate a missing derivation and identify recorded passages affected by a correction. The structure earns its place only if these journeys are useful. It also costs time: maps need authoring, changes need reconciliation, and correspondence still needs mathematical judgment. Several accounts that reuse an estimate differently give those costs a purpose. A stable note for one audience, with a few precise references, may need no such additional structure.

The arrangement builds on existing practices. In particular, leanblueprint already connects informal statements with formal declarations and their dependencies. Lean Exposition's emphasis is the maintained relationship between several authored accounts, their selected passages and the distinct evidence behind them. It makes no claim to have invented informal-to-formal links or demonstrated a comparative learning benefit.

The common record centralises claim identity and recorded evidence. The independent formulations still need comparison and maintenance. Separating the annotations from the prose lets each account read as a paper. The papers and maps can be captured as one edition for another reader or application; external source references identify what to consult, although the referenced bytes need not travel with the package. The operating instructions belong in the repository's guides.

The two journeys suggest concrete tests of value. Can a reader recover the needed curvature argument, retain its hypotheses and resume the contradiction? Can an author find and adjudicate the recorded uses of a changed argument without overlooking an optional route? A comparison should give both groups the same sources and careful ordinary references, then measure correct reconstruction and maintenance effort separately. Finding a destination, passing a structural check and explaining why the second chord restores strictness are different observations. The drafts have received source-based agent review; independent human acceptance and improved comprehension remain unestablished. The intended result is a shorter route from "I do not understand this step" to the argument that resolves it, with its assumptions and credit still attached.

Figure credit. The diagrams adapt this companion's shared-claim, two-input and passage-to-source studies, derived from Will Cook's *A Repository-Based System for Research and Publication* (CC BY 4.0). Their subjects, labels and reading order have been revised for this account.